PHARMACOGNOSY & PHYTOCHEMISTRY

Comprehensive Exam Preparation Question Bank: Glycosides, Tannins & Carbohydrates

1. GLYCOSIDES

1. Stas-otta process is used for extraction of:

a) Glycosides

b) Alkaloids

c) Terpenoids

d) Resins

Ans: a

> **Explanation:** The Stas-Otto process is a classical extraction and purification protocol traditionally used to isolate plant glycosides and alkaloids from complex biological matrices, utilizing selective solubility in water and organic solvents across adjusted pH ranges.

Afaan Oromoo: *Adeemsi 'Stas-otto' jedhamu dhangala'aa keessaa 'Glycosides' addaan baasanii baasuuf tajaajila.*

2. Senna mainly contains:

a) O-glycosides

b) N-glycosides

c) C-glycosides

d) S-glycosides

Ans: a

> **Explanation:** Senna leaves contain anthraquinone derivatives known as sennosides A and B. These are specialized O-glycosides where the sugar moiety is attached to an oxygen atom of the aglycone (anthrone/anthraquinone nucleus).

Afaan Oromoo: *Baalli Senna keessattuu 'O-glycosides' of keessaa qaba; kunis yeroo sukkaarri oksijiinii aglycone irratti hidhamudha.*

3. Senna leaf is under the class of glycoside:

a) Cardiac

b) Cyanogenic

c) Anthracene

d) Saponin

Ans: c

> **Explanation:** Senna belongs to the anthracene or anthraquinone class of glycosides, which are highly recognized for their stimulant laxative properties that act directly on the smooth muscle of the large intestine.

Afaan Oromoo: *Baalli Senna ramaddii glycoside 'Anthracene' jalatti ramadama; kunis garaachaa fi mar'umaan keessatti akka bofantee (laxative) tajaajila.*

4. Which drug is under the chemical class of cyanogenetic glycoside?

a) Bitter almond

b) Black mustard

c) Digitalis

d) Rhubarb

Ans: a

> **Explanation:** Bitter almond (Prunus dulcis var. amara) contains amygdalin, a well-known cyanogenetic glycoside that releases hydrocyanic acid (HCN), benzaldehyde, and glucose upon enzymatic hydrolysis.

Afaan Oromoo: *Qorichi 'Bitter almond' jedhamu ramaddii 'cyanogenetic glycoside' jalatti argama; kunis yeroo dhangala'u summii 'hydrocyanic acid' gadhiisa.*

5. Drug under the class of isothiocyanate glycosides:

a) Black mustard

b) Senega

c) Thevetia

d) Aloe

Ans: a

> **Explanation:** Black mustard (Sinapis nigra) contains sinigrin, which belongs to the isothiocyanate class of glycosides (also known as glucosinolates). Hydrolysis of sinigrin by the enzyme myrosin yields pungent allyl isothiocyanate.

Afaan Oromoo: *Qorichi 'isothiocyanate glycosides' jalatti ramadamu 'Black mustard' (Sanaafii Gurraacha) dha; kunis sinigrin of keessaa qaba.*

6. Shatavari is under the chemical class of glycoside:

a) Sterol

b) Saponin

c) Cyanogenetic

d) Isothiocyanate

Ans: b

> **Explanation:** Shatavari (Asparagus racemosus) is rich in steroidal saponin glycosides, commonly referred to as shatavarins (such as Shatavarin I–IV), which are responsible for its immunomodulatory and adaptogenic activities.

Afaan Oromoo: *Shatavari'n gosa glycoside 'Saponin' jedhamu of keessaa qaba; kunis dandeettii ittisa qaamaa dabaluuf gargaara.*

7. Drug is not under the class of cardiac glycoside:

a) Digitalis

b) Thevetia

c) Indian squill

d) Bitter almond

Ans: d

> **Explanation:** Digitalis, Thevetia, and Indian squill all contain steroidal cardiotonic skeletons (cardenolides or bufadienolides) that act on heart muscles. Bitter almond contains a cyanogenetic glycoside (amygdalin) and possesses no cardiac activity.

Afaan Oromoo: *Qorichoota armaan gadii keessaa kan harki onnee (cardiac glycoside) keessa hin jirre 'Bitter almond' dha.*

8. The drug showing cardiotonic activity:

a) Thevetia

b) Cochineal

c) Rhubarb

d) Aloe

Ans: a

> **Explanation:** Thevetia (Thevetia peruviana / Yellow Oleander) contains cardiac glycosides such as thevetin A, thevetin B, and peruvoside, which exhibit a positive inotropic effect similar to digitalis.

Afaan Oromoo: *Qorichi dalagaa onnee jabeessuu (cardiotonic activity) argisiisu 'Thevetia' dha.*

9. Which drug is used as diuretic?

a) Quillaia

b) Senega

c) Ginseng

d) Gokhru

Ans: d

> **Explanation:** Gokhru (Tribulus terrestris) contains saponin glycosides and nitrates, which promote increased urine flow, making it a well-established traditional diuretic and nephroprotective agent.

Afaan Oromoo: *Qorichi fincaan akka baay'atu gochuuf (diuretic) tajaajilu 'Gokhru' jedhama.*

10. The drug used as nerve tonic:

a) Brahmi

b) Mordica

c) Ginseng

d) Senega

Ans: a

> **Explanation:** Brahmi (Bacopa monnieri) contains triterpenoid saponins known as bacosides, which have documented memory-enhancing, cognitive-pacing, and neuroprotective properties, classifying it as a premier nerve tonic.

Afaan Oromoo: *Qorichi dandeettii sammuu fi narvii jabeessuuf (nerve tonic) tajaajilu 'Brahmi' dha.*

11. The family of Gokhru:

a) Liliaceae

b) Cucurbitaceae

c) Zygophyllaceae

d) Araliaceae

Ans: c

> **Explanation:** Gokhru is botanically derived from the annual herb Tribulus terrestris, which belongs fundamentally to the family Zygophyllaceae.

Afaan Oromoo: *Maatiin biqiltuu Gokhru 'Zygophyllaceae' jedhama.*

12. Borntrager test is used for one of the following drugs:

a) Senna

b) Aloe

c) Digitalis

d) Stropanthus

Ans: a

> **Explanation:** The Borntrager test is a specific color identification reaction used to identify free anthraquinone aglycones. When senna is treated with an acid and an organic solvent followed by an alkali, a distinctive pink, red, or violet color develops in the ammoniacal phase.

Afaan Oromoo: *Qorannoon 'Borntrager test' jedhamu qoricha 'Senna' addaan baasuuf tajaajila; halluu diimaa ykn lila kenna.*

13. In klung's isobarbaloin test, Curacao aloes shows colour:

a) Yellow

b) Blue

c) Wine red

d) Green

Ans: c

> **Explanation:** Klung's isobarbaloin test is used to differentiate varieties of aloe. Curacao aloe contains high amounts of isobarbaloin, which produces a deep wine-red coloration when treated with copper sulfate, sodium chloride, and alcohol.

Afaan Oromoo: *Qorannoo 'Klung's isobarbaloin' irratti, 'Curacao aloes' halluu diimaa daadhii wayinii (wine red) agarsiisa.*

14. In klung's isobarbaloin test, Socotrine and Zanziber aloes show colour:

a) Red

b) Faint yellow

c) Green

d) No colouration

Ans: d

> **Explanation:** Socotrine and Zanzibar aloes contain negligible amounts of isobarbaloin compared to Curacao aloes, resulting in a negative reaction (no coloration) under the specific conditions of Klung's test.

Afaan Oromoo: *Qorannoo 'Klung's' irratti, gosi 'Socotrine' fi 'Zanziber aloes' halluu hin jijjiiran (no coloration).*

15. Alcoholic extract of aloe under UV light gives colour:

a) Blue

b) Red

c) Deep brown

d) Pink

Ans: c

> **Explanation:** Under ultraviolet radiation, alcoholic solutions of diverse aloe species exhibit deep brown to greenish-brown fluorescence because of the core anthraquinone components like barbaloin and aloe-emodin.

Afaan Oromoo: *Cuunfaa Aloe ifa 'UV' jalatti halluu magariisa-bunaa dafee (deep brown) kenna.*

16. The substitute for aloes:

a) Cape aloes

b) Socotrine aloes

c) Curacao aloes

d) Natal aloes

Ans: d

> **Explanation:** Natal aloes (derived from Aloe candelabrum) has historical usage as an economic substitute or adulterant for standard pharmaceutical aloes (Curacao or Cape varieties), although it contains different chemical markers like nataloin.

Afaan Oromoo: *Bakka bu'aan ykn hanni 'Aloe' seeraan alaa 'Natal aloes' jedhama.*

17. The biological source for cape aloes:

a) Curacao aloes

b) Socotrine aloes

c) Aloe ferox

d) Zanziber aloes

Ans: c

> **Explanation:** Cape aloes is prepared from the dried juice of the leaves of Aloe ferox Miller and its hybrids, a plant species indigenous to South Africa.

Afaan Oromoo: *Maddi baayoloojii 'Cape aloes' biqiltuu 'Aloe ferox' jedhamu irraayi.*

18. Saponin glycoside shows one of the following property:

a) Laxative

b) Anticonvulsant

c) Foaming

d) Astringent

Ans: c

> **Explanation:** Saponins lower the surface tension of aqueous solutions, which gives rise to a characteristic, persistent frothing or foaming action when shaken with water. They also possess hemolytic capabilities.

Afaan Oromoo: *Gosi glycoside 'Saponin' jedhamu dandeettii hojjaa/hoofee (foaming) uumuu qaba yeroo bishaan keessatti raasamu.*

19. Out of the following, one is not the example of Cardinolide:

a) Digitoxin

b) Digoxin

c) Gitoxigenin

d) Scillarin A

Ans: d

> **Explanation:** Digitoxin, Digoxin, and Gitoxigenin are cardenolides (5-membered lactone ring attached at C-17). Scillarin A, isolated from squill, is a bufadienolide, featuring a 6-membered unsaturated lactone ring at the C-17 position.

Afaan Oromoo: *Fakkeenya qoricha 'Cardinolide' kan hin taane 'Scillarin A' dha; inni bufadienolide dha.*

20. The biological source for Dioscoria:

a) dioscoria deltoid

b) dioscoria floribunda

c) dioscoria villosa

d) Dioscorea compositae

Ans: a

> **Explanation:** Dioscorea (Yam) is commercially obtained from the dried rhizomes of Dioscorea deltoidea Wall., Dioscorea composita, or other species, yielding rich quantities of the steroidal sapogenin diosgenin.

Afaan Oromoo: *Maddi guddaa fi seeraa kan biqiltuu 'Dioscoria' 'Dioscoria deltoidea' jedhama.*

21. Yam is the synonym of the drug:

a) Stropanthus

b) Dioscoria

c) Safed musali

d) Liquorice

Ans: b

> **Explanation:** Dioscorea species are widely known by the common vernacular name "Yam" or "Wild Yam," recognized globally for their high content of steroidal precursors.

Afaan Oromoo: *Maqaan biraa kan qoricha 'Dioscoria' jedhamu 'Yam' jedhama.*

22. Dioscoria is used in the treatment of:

a) Ulcer

b) Cancer

c) Rheumatic arthritis

d) Kidney stone

Ans: c

> **Explanation:** Dioscorea yields diosgenin, which is a structural precursor for the industrial semi-synthesis of systemic corticosteroids, oral contraceptives, and sex hormones used medically to manage chronic inflammation like rheumatoid arthritis.

Afaan Oromoo: *Biqiltuun Dioscoria dhukkuba lafee fi buusaa (Rheumatic arthritis) wal'aanuuf tajaajila.*

23. _____ Is not the synonym of Liquorice.

a) Glycyrrhiza

b) Liquorice root

c) Mulethi

d) Yam

Ans: d

> **Explanation:** Glycyrrhiza, Liquorice root, and Mulethi are direct synonyms for Glycyrrhiza glabra. Yam is the synonym for Dioscorea, an entirely different taxonomic entity.

Afaan Oromoo: *Maqaa armaan gadii keessaa kan Liquorice hin ibsine 'Yam' dha; inni kan Dioscoria ti.*

24. Liquorice belongs to family:

a) Liliaceae

b) Triterpenoid

c) Loganaceae

d) Leguminosae

Ans: d

> **Explanation:** Liquorice (Glycyrrhiza glabra) belongs to the family Leguminosae (Fabaceae), characterized by its distinctive legume-like or pod-bearing structures.

Afaan Oromoo: *Maatiin biqiltuu Liquorice 'Leguminosae' (gosa akaayii) jedhama.*

25. Glycyrrhizinic acid on hydrolysis gives:

a) Glycyrrhetic acid

b) Glycyrrhizin

c) Liquiritin

d) Isoliquiritin

Ans: a

> **Explanation:** Acid or enzymatic hydrolysis of glycyrrhizin (glycyrrhizinic acid) splits off two molecules of D-glucuronic acid, leaving behind the triterpenoid aglycone core known as glycyrrhetic acid (glycyrrhetinic acid).

Afaan Oromoo: *'Glycyrrhizinic acid' yeroo bishaan keessatti dhangala'u asidii 'Glycyrrhetic acid' jedhamu kenna.*

26. Liquorice is used in the treatment of:

a) Conjuctivitis

b) Peptic ulcer

c) Allergy

d) Skin disease

Ans: b

> **Explanation:** Liquorice and its deglycyrrhizinated forms (DGL) promote healing of the gastric mucosa by increasing mucous synthesis and prostaglandins, making it highly effective against gastric and peptic ulcers.

Afaan Oromoo: *Liquorice dhukkuba madaa garaachaa (Peptic ulcer) wal'aanuuf baay'ee gargaara.*

27. When glycyrrhiza is treated with 80% sulphuric acid, then it shows colour:

a) Yellow

b) Red

c) Green

d) Violet

Ans: a

> **Explanation:** Treating a clear cross-section or powder of liquorice root containing glycyrrhizin directly with 80% v/v sulfuric acid produces a prominent yellow-orange color due to chromogenic responses of the underlying triterpenoids.

Afaan Oromoo: *Yeroo biqiltuun kun asidii 'sulphuric acid' (80%) tiin tajaajilamu halluu keelloo (yellow) agarsiisa.*

28. Rhubarb is cultivated mainly in which state of India:

a) J and K

b) Madhya Pradesh

c) Maharashtra

d) Bengal

Ans: a

> **Explanation:** Indian Rhubarb (Rheum emodi / Rheum webbianum) grows naturally and is cultivated in the high-altitude Himalayan tracts of Jammu & Kashmir and Himachal Pradesh.

Afaan Oromoo: *Biqiltuun Rhubarb jedhamu kun keessattuu biyya Hindii bulchiinsa 'Jammu and Kashmir' (J & K) keessatti biqila.*

29. Rhubarb when treated with alkali, shows red colour due to presence of:

a) Anthroquionoe glycoside

b) Cardiac glycoside

c) sapogenin glycoside

d) Cyanogenic glycoside

Ans: a

> **Explanation:** Rhubarb contains substantial quantities of free and bound anthraquinones (like emodin, chrysophanol, rhein). When exposed to alkaline conditions, these undergo resonance stabilization changes to give a brilliant red color.

Afaan Oromoo: *Rhubarb yeroo 'alkali' dhandhamu halluu diimaa kenna; kunis sababa 'Anthraquinone glycoside' of keessaa qabuufi.*

30. Keller kiliani test is positive for:

a) Digitoxose

b) Gitoxose

c) digitoxigenin

d) gitoxigenin

Ans: a

> **Explanation:** The Keller-Kiliani test is specific for the detection of 2-deoxy-hexose sugars, such as digitoxose, which are fundamental carbohydrate monomers present in cardiac glycosides from digitalis.

Afaan Oromoo: *Qorannoon 'Keller kiliani' sukkaara 'Digitoxose' jedhamuuf qofa 'positive' ta'a.*

31. Presence of star spots is the characteristic of drug:

a) Cascara

b) Senna

c) Aloe

d) Rhubarb

Ans: d

> **Explanation:** Morphologically, the transverse section of official Rhubarb (Rheum) displays characteristic anomalous fibrovascular bundles located within the pith region, widely known as "star spots."

Afaan Oromoo: *Mallattoon 'star spots' (faana urjii) keessoo biqiltuu 'Rhubarb' irratti addatti calaqqisa.*

32. Modified Borntroger test is positive for drug:

a) Aloe

b) Senna

c) Digitalis

d) Senna pod

Ans: a

> **Explanation:** The standard Borntrager test works only on free anthraquinones (O-glycosides after hydrolysis). Aloe contains C-glycosides (like aloin) where the carbon-carbon bond is highly stable. The modified test utilizes ferric chloride ($FeCl_3$) to oxidatively cleave the C-C bond, enabling a positive pink-red result.

Afaan Oromoo: *Qorannoon 'Modified Borntroger test' jedhamu qoricha 'Aloe' tiif mirkanaa'aa (positive) ta'a; sababni isaas 'C-glycosides' waan qabuuf.*

33. Botanical source of Arjuna bark:

a) Terminalis tamentosa

b) Polygala chinesis

c) Poligala erioptera

d) Terminalis arjuna

Ans: d

> **Explanation:** Arjuna bark is obtained from the dried stem bark of Terminalia arjuna Bedd., belonging to the family Combretaceae, and is heavily used as a traditional cardiotonic.

Afaan Oromoo: *Maddi biqiltuu gogaa 'Arjuna' 'Terminalia arjuna' jedhama.*

34. Presence of aleurone grain is the characteristic of the drug:

a) Belladonna

b) Nux vomica

c) Hyoscyamus

d) Vinca

Ans: b

> **Explanation:** The cells of the dense, horny endosperm of Nux vomica seeds are filled with fixed oils and large, explicit protein storage deposits called aleurone grains.

Afaan Oromoo: *'Aleurone grain' (gosa pirootinii) keessattuu sanyii 'Nux vomica' keessatti argama.*

35. Drug with the presence of plasmodesma:

a) Nux vomica

b) Rauwolfia

c) Vinca rosea

d) Phsysostigma

Ans: a

> **Explanation:** Microscopic evaluation of Nux vomica seed endosperm reveals highly specialized inter-cellular protoplasmic bridges extending through cell wall channels, structurally identified as plasmodesmata.

Afaan Oromoo: *Hidhaan tishuu 'Plasmodesma' jedhamu sanyii 'Nux vomica' keessatti caasaa microscopic tiin mul'ata.*

36. Digitalis leaves should be dried at temperature below:

a) 20 C

b) 30 C

c) 40 C

d) 60 C

Ans: d

> **Explanation:** Digitalis leaves must be dried rapidly at a temperature below 60°C immediately after collection to halt endogenous glycosidases (enzymes) from breaking down the medically valuable primary cardiac glycosides into secondary forms.

Afaan Oromoo: *Baalli 'Digitalis' ho'a digrii 60 C gadi ta'een gogfamuu qaba; kunis akka 'enzymes' qoricha sana hin balleessineefidha.*

37. Vein-islet no. is Indian senna is in the range of:

a) 10.5 to 12.5

b) 13.5 to 16.5

c) 16.5 to 18.0

d) 19.5 to 22.5

Ans: d

> **Explanation:** Vein-islet number represents the number of vein islets per square millimeter of leaf surface. In Indian Senna (Cassia angustifolia), this quantitative evaluation parameter ranges characteristically from 19.5 to 22.5.

Afaan Oromoo: *Lakkoofsi 'Vein-islet' baala Senna Hindii (Indian senna) giddugaleessaan 19.5 hanga 22.5 ta'a.*

38. Stomatal Index in Indian senna is in the range of:

a) 17 to 20

b) 20 to 23

c) 23 to 26

d) 26 to 29

Ans: a

> **Explanation:** Stomatal index is the percentage percentage of stomata relative to the total epidermal cells. For Indian senna leaves, the standard value falls reliably between 17 and 20.

Afaan Oromoo: *'Stomatal Index' baala Senna Hindii giddugaleessaan 17 hanga 20 gidduutti argama.*

39. Palisade ratio in Alexandrian senna is:

a) 7

b) 10

c) 13

d) 16

Ans: a

> **Explanation:** Palisade ratio is the average number of palisade cells located beneath each upper epidermal cell. For Alexandrian Senna (Cassia acutifolia), this value is consistently lower, averaging around 7.

Afaan Oromoo: *Reeshoon 'Palisade' baala Senna Istiriiriyaa (Alexandrian senna) irratti 7 dha.*

40. Vein- islet no. in Alexandrian senna is in the range of:

a) 10 to 15.5

b) 16 to 20.5

c) 25 to 29.5

d) 20 to 24.5

Ans: c

> **Explanation:** Alexandrian senna exhibits a structurally distinct and denser network of small veins compared to Indian senna, raising its quantitative vein-islet count to a range of 25 to 29.5.

Afaan Oromoo: *Lakkoofsi 'Vein-islet' baala Senna Alexandrian dhiphoo waan ta'eef 25 hanga 29.5 gidduutti argama.*

Glycosides - Interactive Matching Matrix

Drug Parameter	Correct Association / Value	Phytochemical Domain
Indian Senna Family	Leguminosae	Anthracene Class
Aloe Family	Liliaceae	C-Glycoside Class
Rhubarb Family	Polygonaceae	Anthraquinone Class
Cascara Family	Rhamnaceae	Purgative Action
Cyanogenetic Target	Bitter Almond (Amygdalin)	Enzyme Hydrolysis
Red Squill Activity	Rat Poisoning (Rodenticide)	Bufadienolide Class
Brahmi Action	Nervine Tonic / Cognitive	Saponin Tracing

2. TANNINS

1. Goldbeater skin test is used for identification of:

a) Alkaloids

b) Glycosides

c) Tannins

d) Resins

Ans: c

> **Explanation:** The Goldbeater skin test is the gold standard for verifying true tannins. An animal membrane (ox skin) is treated with the sample followed by ferrous sulfate; true tannins bind tightly to the membrane proteins and produce a permanent black or dark brown color.

Afaan Oromoo: *Qorannoon 'Goldbeater skin' jedhamu gosa qoricha 'Tannins' addaan baasuuf gargaara; gogaa loonii irratti halluu gurraacha fida.*

2. Tannins show some chemical reaction except:

a) Solution of tannins precipitate gelatin

b) Tannins are precipitated by salt of copper, tin and lead

c) They show colour reaction with iron

d) They react with potassium mercuric iodide

Ans: d

> **Explanation:** Potassium mercuric iodide is Mayer's reagent, which precipitates nitrogenous alkaloidal bases. Tannins precipitate gelatin and heavy metals (copper, lead) and react with iron compounds, but do not react positively with Mayer's reagent.

Afaan Oromoo: *Tannins amaloota armaan gadii hunda agarsiisa, garuu 'potassium mercuric iodide' (Mayer's reagent) wajjin wal-hin fudhatu (kun kan alkaloid ti).*

3. Condensed tannins are called as:

a) Hydrolysable tannins

b) Non hydrolysable

c) Pseudotannins

d) Prototannins

Ans: b

> **Explanation:** Condensed tannins are polymers of flavan-3-ols (catechins) or flavan-3,4-diols (leucoanthocyanidins) linked by carbon-carbon bonds. Unlike hydrolysable tannins, they resist simple hydrolytic breakdown and are termed non-hydrolysable tannins.

Afaan Oromoo: *'Condensed tannins' jechuun gosoota bishaan keessatti salphaatti hin dhangalaane (Non hydrolysable) jechuudha.*

4. Tannic acid is under the class of:

a) Condensed tannins

b) Pseudotannins

c) Hydrolysable tannins

d) Condensed and pseudotannins

Ans: c

> **Explanation:** Tannic acid belongs to the hydrolysable class of tannins. Upon treatment with dilute mineral acids or tannase enzymes, it breaks down into simpler phenolic acids (such as gallic acid) and a central sugar molecule (glucose).

Afaan Oromoo: *'Tannic acid' asidii gosa 'Hydrolysable tannins' jalatti argamudha; salphaatti gara gallic acid tii jijjiirama.*

5. Biological source of Arjuna:

a) Terminalia arjuna

b) Terminalia chebula

c) Terminalia tomenstosa

d) Terminalia belerica

Ans: a

> **Explanation:** Arjuna is derived from the dried stem bark of the tree Terminalia arjuna Wight & Arn., belonging to the family Combretaceae.

Afaan Oromoo: *Maddi seeraa kan Arjuna 'Terminalia arjuna' jedhama; gogaa mukichaa irraa argama.*

6. Tannic acid is antidote in poisoning:

a) Iodine

b) Cyanide

c) Alkaloid

d) Arsenic

Ans: c

> **Explanation:** Tannic acid readily forms highly insoluble complexes and chemical precipitates when mixed with alkaline, nitrogenous alkaloids. Administered orally, it prevents the systemic absorption of toxic alkaloids from the gastrointestinal tract.

Afaan Oromoo: *'Tannic acid' summii gosa 'Alkaloid' ta'an qaama keessatti dafanii akka hin absorbed taane ittisuuf qoricha farra summii (antidote) ta'ee tajaajila.*

7. When pterocarpus is treated with potassium hydroxide, it gives:

a) Violet

b) Green

c) Red

d) Black

Ans: a

> **Explanation:** Pterocarpus (Pterocarpus marsupium / Kino) contains kinotannic acid. Treatment of its aqueous or alcoholic extract with potassium hydroxide (KOH) yields a dramatic violet color change due to rapid phenolic oxidation.

Afaan Oromoo: *Pterocarpus yeroo 'potassium hydroxide' wajjin tajaajilamu halluu dhiilgee ykn lila (violet) kenna.*

8. Gambir flurescin test is used for drug:

a) Arjuna

b) Ashoka

c) Pale catechu

d) Black catechu

Ans: c

> **Explanation:** Pale catechu (Uncaria gambier) contains gambir-fluorescein. When extracted with alcohol and treated with sodium hydroxide followed by petroleum ether, the ether layer exhibits a brilliant, distinctive green fluorescence. Black catechu does not show this property.

Afaan Oromoo: *Qorannoon 'Gambir fluorescein' jedhamu qoricha 'Pale catechu' addaan baasuuf gargaara; ifa calaqqisaa magarii uuma.*

9. Black catechu contains about percentage of acacatechin:

a) One percentage

b) Three percent

c) Seven percent

d) Ten percent

Ans: d

> **Explanation:** Black catechu (Acacia catechu) contains roughly 2% to 10% of acacatechin, a characteristic crystalline flavenol compound along with catechu-tannic acid.

Afaan Oromoo: *'Black catechu' keessatti hamma 'acacatechin' argamu dhibbeentaa kudhan (10%) qabata.*

3. CARBOHYDRATES

1. Composition of Molisch reagent is:

a) Alpha napthol + sulphuric acid

b) Beta napthol + sulphuric acid

c) Naphthylamine + sulphuric acid

d) Phenol + sulphuric acid

Ans: a

> **Explanation:** Molisch's reagent is a 5% or 10% solution of $ lpha$-naphthol dissolved in ethanol. When added to a carbohydrate solution along with concentrated sulfuric acid (H_2SO_4), dehydration of the sugar yields furfural derivatives that condense with $ lpha$-naphthol to create a violet or purple ring.

Afaan Oromoo: *Qorichi 'Molisch reagent' jedhamu 'Alpha napthol' fi 'sulphuric acid' irraa tolfama; sukkaara kamiyyuu mirkaneessuuf gargaara.*

2. Chemical test used for identification of carbohydrates:

a) Ninhydrin

b) Molisch

c) Borntrager

d) Baljet

Ans: b

> **Explanation:** The Molisch test is the general group test used to screen for the presence of any carbohydrate material. Ninhydrin detects amino acids; Borntrager screens for anthraquinones; Baljet tests for cardiac lactones.

Afaan Oromoo: *Qorannoon kiimikaalaa kaarboohaydreetii (carbohydrates) mirkaneessuuf gargaaru 'Molisch test' jedhama.*

3. Important chemical constituent of Algin is:

a) Tragacanthin

b) Bassorin

c) Alginic acid

d) Alginic ester

Ans: c

> **Explanation:** Algin (sodium alginate), obtained from brown seaweeds, is chemically composed of the salt forms of alginic acid, which is a linear hydrophilic polymer composed of poly-$ eta$-D-mannuronic acid and poly-$ lpha$-L-guluronic acid units.

Afaan Oromoo: *Waanti kiimikaalaa guddaan 'Algin' keessatti argamu 'Alginic acid' jedhama; saansa bishaanii keessaa argama.*

4. Important chemical constituent honey is:

a) Glucose

b) Fructose

c) Maltose

d) Invert sugar

Ans: d

> **Explanation:** Honey consists predominantly of a natural mixture of D-fructose and D-glucose, structurally termed "invert sugar" (approx. 70-80%), which is formed via the enzymatic inversion of sucrose by honeybees.

Afaan Oromoo: *Sukkaarri guddaan damma keessatti argamu 'Invert sugar' jedhama; kunis walitti dhufeenya fructose fi glucose ti.*

5. Pectin is used in the pharmaceutical industries as:

a) Adsorbent

b) Demulcent

c) Emulsifying agent

d) Suspending agent

Ans: a

> **Explanation:** Pectin possesses unique macromolecular colloidal structures that allow it to act as an effective gastrointestinal adsorbent of toxins and bacterial byproducts, traditionally utilized to formulate anti-diarrheal mixtures.

Afaan Oromoo: *'Pectin' indastri qorichaa keessatti akka summii fi baakteeriyaa xuuxuuf (Adsorbent) tajaajila; baala madaa quufsuuf gargaara.*

6. Liquid glucose is obtained by:

a) Partial hydrolysis of glucose

b) Partial hydrolysis of starch

c) Complete hydrolysis of starch

d) Complete hydrolysis of glucose

Ans: b

> **Explanation:** Liquid glucose is industrially manufactured through the regulated, incomplete (partial) acid or enzymatic hydrolysis of purified corn starch, yielding a viscous mixture of glucose, maltose, and dextrins.

Afaan Oromoo: *Glucose dhangala'aan (Liquid glucose) kan argamu 'Starch' (istarchii) xannacha bishaaniitiin hamiisa walakkaa (partial hydrolysis) yeroo ta'udha.*

7. The biological source for Indian gum is:

a) Cyamopsis tetragonoglobules

b) Astragalus gamifer

c) Acacia Arabica

d) Acacia Senegal

Ans: c

> **Explanation:** Indian gum (Gum Acacia) is obtained as a dried gummy exudate from the stem and branches of Acacia arabica Willd. (also known as Acacia nilotica), belonging to the family Leguminosae.

Afaan Oromoo: *Maddi seeraa baayoloojii kan haphee Hindii (Indian gum) 'Acacia arabica' jedhamu irraayi.*

8. When Indian gum is treated with ruthenium red' it shows:

a) Pink colour

b) Red colour

c) Blue

d) No colour change

Ans: d

> **Explanation:** Ruthenium red stains mucilages and gums containing free uronic acid units (like agar or pectin) pink. True Indian Gum (Acacia arabica) contains highly branched arabinogalactan complexes that remain completely unstained (no color change) by ruthenium red.

Afaan Oromoo: *Yeroo hapheen Hindii qoricha 'ruthenium red' tiin tajaajilamu halluu hin jijjiiru (no colour change).*

9. When Indian gum is treated with hydrogen peroxide and benzidine in alcohol; is shows blue colour due to:

a) Hydrolyase enzyme

b) Oxidase enzyme

c) Reductase enzyme

d) Invertase enzyme

Ans: b

> **Explanation:** Crude Gum Acacia naturally contains an active, endogenous peroxidase/oxidase enzyme system. In the presence of hydrogen peroxide (H_2O_2), this enzyme oxidizes benzidine into a highly conjugated blue complex.

Afaan Oromoo: *Hapheen Hindii halluu cuquliisa kan agarsiisu sababa 'Oxidase enzyme' keessa jiruun 'benzidine' waan addeessuufi.*

10. The mostly used adulterant for Indian gum is:

a) Acacia Senegal gum

b) Gum from Astragalus gummifer

c) Gum ghatti from anogeissus latifolia

d) Guar gum

Ans: c

> **Explanation:** Gum ghatti, an exudate harvested from Anogeissus latifolia, is commonly utilized as a commercial substitute or adulterant for authentic Indian Gum owing to similar physical viscosities and appearances.

Afaan Oromoo: *Haphee Hindii waliin waanti hanni/sobaa baay'ee makamu 'Gum ghatti' (kan mukaa Anogeissus latifolia) ti.*

11. Synonym for Guar gum:

a) Jaguar gum

b) Acacia

c) Gum acacia

d) Gum Arabica

Ans: a

> **Explanation:** Guar gum, derived from the seeds of Cyamopsis tetragonoloba, is referred to in various industrial settings by the trade name "Jaguar gum."

Afaan Oromoo: *Maqaan biraa kan 'Guar gum' jedhamu 'Jaguar gum' jedhama.*

12. When guar gum is treated with about 2% solution of lead acetate; it shows:

a) Pink colour

b) Blue colour

c) Yellow colour

d) No colour change

Ans: d

> **Explanation:** Lead acetate separates out specific gums by precipitating them. Unlike standard plant mucilages, guar gum is a neutral galactomannan polymer and does not form an instant precipitate or color change with a 2% lead acetate solution.

Afaan Oromoo: *Guar gum yeroo dhangala'aa 'lead acetate' (2%) tiin tajaajilamu halluu hin jijjiiru (no colour change).*

13. Artificial invert sugar is an adulterant for honey and it is detected by the test:

a) Tollen’s

b) Ninhydrine

c) Baljet test

d) Fiehe’s test

Ans: d

> **Explanation:** Adulterating honey with cheap, acid-hydrolyzed artificial invert sugar leaves trace markers of 5-hydroxymethylfurfural (HMF). Fiehe’s test detects this compound via a vivid red color reaction when treated with resorcinol in hydrochloric acid.

Afaan Oromoo: *Sukkaarri sobaa damma keessatti makame qorannoo 'Fiehe's test' jedhamuun qofaan saaxilama (halluu diimaa kenna).*

14. The water soluble portion of tragacanth is known as:

a) Tragacanthin

b) Bassorin

c) Galctouronic acid

d) D-galctopyranose

Ans: a

> **Explanation:** Natural Gum Tragacanth separates into two major fractions: Tragacanthin, which dissolves readily in water to form a clear colloidal sol, and Bassorin, which swells extensively in water but remains insoluble.

Afaan Oromoo: *Kutaan biqiltuu Tragacanth bishaan keessatti baay'ee dhangala'u (water soluble) 'Tragacanthin' jedhama.*

15. Pectin is used in the treatment of:

a) Ulcer

b) Hypertension

c) Angina

d) Diarrhoea

Ans: d

> **Explanation:** Pectin acts as an outstanding demulcent and heavy industrial adsorbent within the intestinal lumen, widely indicated to manage acute diarrhea and restore solid fecal consistency.

Afaan Oromoo: *Pectin keessattuu dhukkuba garaa kaasaa (Diarrhoea) wal'aanuuf gargaara.*

16. Biological source for Isapgol:

a) Plantago ovata

b) Aegle marmelos

c) Giladinium amansii

d) Condrus cripsus

Ans: a

> **Explanation:** Isapgol (Psyllium) consists of the dried seeds or cleaned husks stripped from the plant Plantago ovata Forsskal, belonging to the family Plantaginaceae.

Afaan Oromoo: *Maddi guddaa baayoloojii kan qoricha Isapgol 'Plantago ovata' jedhamu irraayi.*

17. Which test is used for the purity of Isapgol?

a) Barfoed test

b) Millon’s test

c) Molisch test

d) Swelling factor

Ans: d

> **Explanation:** The purity and mucilage-yielding capacity of Isapgol seeds and husks are measured directly by its "swelling factor"—the volume in milliliters occupied by 1 gram of the drug after swelling in water for 24 hours.

Afaan Oromoo: *Qulqullina sanyii Isapgol mirkaneessuuf qorannoo 'Swelling factor' (hamma iitahuu bishaan keessatti) jedhamutti tajaajilamu.*

18. When Isapgol is treated with rhuthenium red, it shows colour:

a) Blue

b) Yellow

c) Pink

d) Green

Ans: c

> **Explanation:** The outer epidermis of Isapgol seeds contains thick layers of acidic mucilage. This complex picks up the ruthenium red stain, producing a bright pink color under microscopic inspection.

Afaan Oromoo: *Isapgol yeroo 'ruthenium red' tiin tajaajilamu halluu diimaa dhiilgee (pink) kenna.*

19. Isapgol seeds are adulterated with:

a) Plantago purshii

b) Plantago lanciolata

c) Plantago aristala

d) Plantago pysllium

Ans: b

> **Explanation:** Seeds of Plantago lanceolata (wild psyllium) serve as common geographic adulterants for official Plantago ovata seeds, but can be distinguished by their lower swelling capacity.

Afaan Oromoo: *Sanyiin Isapgol keessattuu sanyii daggalaa kan 'Plantago lanceolata' jedhamuun makamee hatama.*

20. Ager is used as:

a) Binder

b) Disintigrant

c) Emulsifying agent

d) Preservative

Ans: c

> **Explanation:** In pharmacy, Agar is widely used as a hydrocolloid binding matrix, a thickening agent, and a stable emulsifying agent for mineral oil preparations, as well as a solidifying base for microbial agar plates.

Afaan Oromoo: *Agar indastri qorichaa keessatti akka jabaataa zayitaa fi bishaan walitti maku (Emulsifying agent) tajaajila.*

21. Amylum is the synonym for:

a) Tragacanth

b) Starch

c) Inulin

d) Locust bean

Ans: b

> **Explanation:** "Amylum" is the classical Latin pharmaceutical term designated specifically for commercial starches sourced from corn, wheat, rice, or potatoes.

Afaan Oromoo: *Maqaan Lootinii kan 'Starch' (istarchii) ibsu 'Amylum' jedhama.*

22. Rice starch is of size:

a) 2-12 micron

b) 14-17 micron

c) 17-20 micron

d) 20-25 micron

Ans: a

> **Explanation:** Rice starch grains (Oryza sativa) are known for being exceptionally small. They typically present as tiny, angular polyhedral structures measuring between 2 and 12 micrometers (μm) in diameter.

Afaan Oromoo: *Hammamtaan istarchii muxxee (Rice starch) baay'ee xiqqoodha; 2 hanga 12 micron ta'a.*

23. Wheat starch is of size:

a) 0.1 to 1 micron

b) 2 to 3 micron

c) 3 to 5 micron

d) 5 to 50 micron

Ans: d

> **Explanation:** Wheat starch displays a bimodal particle size distribution, containing large, lens-shaped (lenticular) grains ranging from 30 to 50 μm alongside small spherical grains measuring 2 to 10 μm, creating an overall functional range of 5 to 50 μm.

Afaan Oromoo: *Hammamtaan istarchii qamadii (Wheat starch) ballinaan 5 hanga 50 micron gidduutti argama.*

24. The size of potato starch varies from:

a) 30 to 100 micron

b) 100 to 130 micron

c) 130 to 10 micron

d) 160 to 180 micron

Ans: a

> **Explanation:** Potato starch grains (Solanum tuberosum) are among the largest botanical starch configurations, presenting as flattened ovoid granules with an off-center hilum that measure between 30 and 100 μm.

Afaan Oromoo: *Istarchiin dinichaa (Potato starch) gosa baay'ee gurguddaa ta'aniidha; 30 hanga 100 micron deema.*

25. Starch contains amylase and amylopectin in the proportion:

a) 1:2

b) 1:1

c) 1:3

d) 2:1

Ans: a

> **Explanation:** Natural starches contain two main polysaccharides: linear amylose and highly branched amylopectin. In typical grains, these molecules are present in an approximate 1:2 to 1:3 ratio (roughly 20-30% amylose and 70-80% amylopectin).

Afaan Oromoo: *Starch keessatti hamma 'amylase' fi 'amylopectin' reeshoo 1:2 ta'een argamu.*

26. The test, which is not used for identification of carbohydrates:

a) Molisch test

b) Osazone formation test

c) Ninhydrin test

d) Resorcinol test

Ans: c

> **Explanation:** Molisch, Osazone formation, and Resorcinol tests identify various sugars. The Ninhydrin test is selective for the structural $ lpha$-amino acid configurations found in free peptides and proteins.

Afaan Oromoo: *Qorannoo armaan gadii keessaa kan carbohydrates addaan baasuuf hin gargaarre 'Ninhydrin test' dha; innis pirootiniif tajaajila.*

27. Keller kiliani test is used for identification of:

a) Fructose

b) Maltose

c) Deoxy sugar

d) Glucose

Ans: c

> **Explanation:** As noted within the cardiac glycoside parameters, the Keller-Kiliani test is a dedicated chemical color response engineered to detect 2-deoxy sugars (like digitoxose).

Afaan Oromoo: *Qorannoon 'Keller kiliani' gosa sukkaara 'Deoxy sugar' jedhamu addaan baasuuf tajaajila.*

28. The synonym of Lactose is:

a) Milk sugar

b) Butter sugar

c) Crystalline sugar

d) Pure sugar

Ans: a

> **Explanation:** Lactose is a disaccharide containing glucose and galactose units that is found naturally in milk, earning it the universal common synonym "Milk sugar."

Afaan Oromoo: *Maqaan biraa kan Lactose 'Milk sugar' (sukkaara aannanii) jedhama.*

Carbohydrates - Starch & Gum Micro-Evaluation Matrix

Carbohydrate Drug	Biological Sourced Plant	Standard Particle Sizes / Group Class
Carrageenan	Chondrus crispus	Marine Hydrocolloid Algae
Acacia (Indian Gum)	Acacia arabica	Oxidase Positive Exudate
Agar-Agar	Gelidium amansii	Emulsifying Galactan Sol
Gum Ghatti	Anogeissus latifolia	Acacia Standard Substitute
Guar (Jaguar) Gum	Cyamopsis tetragonolobus	Lead Acetate Neutral Complex
Rice Starch (Amylum)	Oryza sativa	Extremely Fine: 2 to 10 um Grains
Potato Starch	Solanum tuberosum	Macroscopic/Large: 30 to 100 um Grains
Wheat Starch	Triticum sativum	Bimodal Distribution: 2 to 45 um Grains